# Syed AbuTalib

Software Engineer | Seattle, WA

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## Experience

**Google** *Seattle, WA*

Software Engineer – Google Chrome (Core Systems & Platform Infrastructure) *2024 - Present*

- Shipped **IAMF** (immersive/spatial audio) decoding in **Chrome 153** to **3B+ users**: built the **C++** streaming decoder and authored the design doc and public **Blink Intent-to-Ship**.
- Found and fixed a **10-year-old integer overflow** in Chromium's `base::CheckedNumeric` safe-math library, used across the codebase; recognized with a **Security Peer Bonus**.
- Re-architected the **C++** multiprocess audio pipeline and **Mojo IPC** interfaces, replacing enums with validated structs to extend layouts from 7.1 to **7.1.4 spatial audio**; earned **OWNERS** status for `//media/audio`.
- Built automated **LibFuzzer/ClusterFuzz** pipelines for the audio stack, finding and fixing **20+ memory-safety bugs** in **AOMedia's** IAMF Decoder; integrated AVFoundation for Chrome on macOS in partnership with **Apple**.

**Google** *Seattle, WA*

Software Engineer – Google Cloud Marketplace & Commerce *June 2022 - 2024*

- Architected an automated Partner VM ingestion and **CVE validation** pipeline in **Java** and **GCP Container Analysis**, cutting review turnaround from **days to under 5 minutes** and saving **5,500 engineering hours annually**.
- Designed and built the full-stack **Terraform Post-Deployment** portal (**TypeScript/Angular** UI, **Java/gRPC** back-end), migrating **all Marketplace VM listings** off legacy Cloud Deployment Manager.
- Authored design docs and built the **Customer Agreements & Entitlements** engine on **Cloud Spanner** and **gRPC/Protobuf**, enforcing real-time procurement guards across VM, GKE, and SaaS.
- Owned and scaled high-availability commerce backend microservices powering the **$4B+ annual Google Cloud Marketplace**, maintaining a **99.99% uptime SLA**.

**Liberty Mutual Insurance** *Boston, MA*

Software Engineer (Private Investments) *Jan. 2021 - June 2022*

- Built a full-stack document management platform with **Django, React, and PostgreSQL**, replacing email-based versioning with a centralized repository to eliminate data collisions.
- Designed a standardized pagination framework across core **REST APIs**, reducing payload sizes and improving **P99 response latencies by 30%** across high-volume investment services.
- Containerized legacy applications with **Docker** and drove **AWS CDK** Infrastructure-as-Code adoption via reusable cloud constructs, **cutting CI/CD build times by 80%**.
- Integrated automated **Jest and Jasmine** testing frameworks into front-end applications, expanding unit and integration coverage and **reducing production regressions by 30%**.

## Skills

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| **Backend & Infra** | Java, Python, C++, Go, Spanner, PostgreSQL, MySQL, Redis, gRPC, Protobuf, REST APIs |
| **Cloud & DevOps** | Google Cloud (GCP, GKE), AWS (CDK), Docker, Kubernetes, Terraform, CI/CD, Linux/Bash |
| **Frontend & Web** | TypeScript, Angular, React, Next.js, Django, Spring Boot, Node.js, Jest, Jasmine |
| **Systems & Security** | Chromium Architecture, Mojo IPC, Multithreading, Memory Safety, LibFuzzer, ClusterFuzz |

## Education

**The University of Texas at Dallas** *Dallas, TX*

Bachelor of Science in Computer Science *Dec. 2020*

- 3.7/4.0 (Cum Laude)